

Realtime System – Exam SS20

Task 1a: Name the 2 most important requirements for real time systems:

- Predictability
- Reliability
- Minimal Delay/Latency

Task 1b: Name 3 types of hardness of real-time systems with short explanation. Give an Example for each

- Hard Realtime: Missing a deadline is a total system failure (airbag in car) Soft Realtime: the usefulness of a result degrades after its deadline, thereby degrading the systems quality of service (warning systems) Firm Realtime: infrequent deadline misses are tolerable but may degrade the systems quality of service. The Usefulness of a result is zero after its deadline. (car: ignition-point-optimizer for motor)

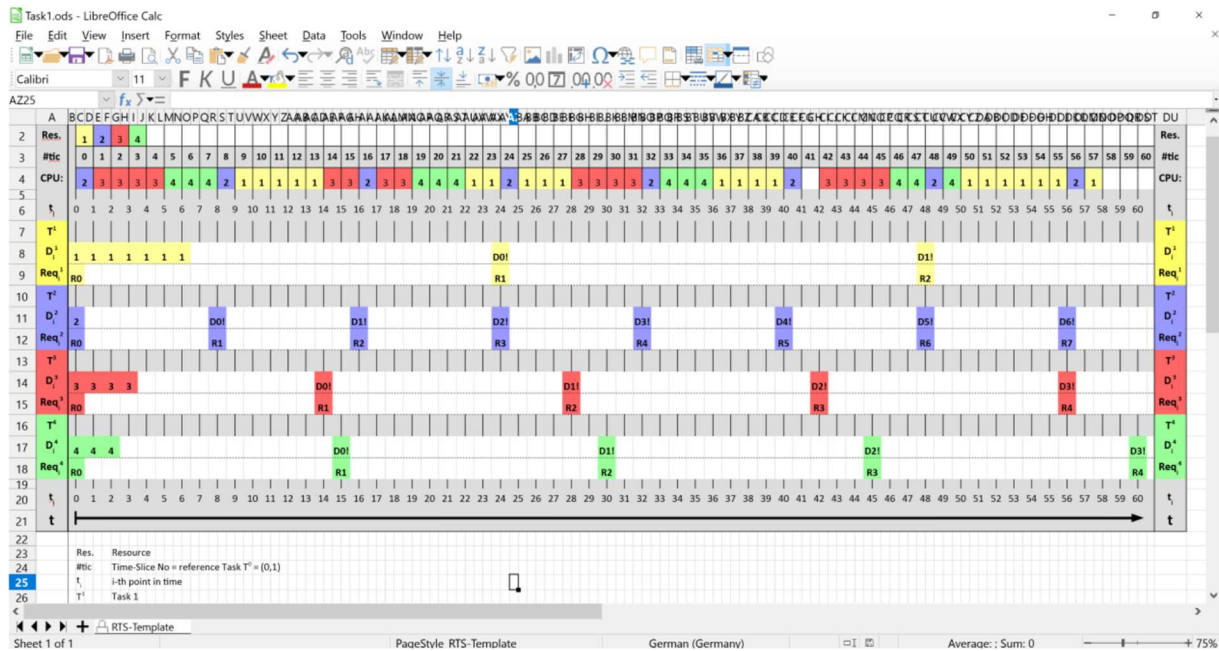
Task 2: Name all Realtime System Tasks:

- Periodic task: When the time of the reappearing is constant, the task is called a periodic task
- Sporadic task: When the occurrence is random but there is a minimum time difference between two jobs of the same task, then the task is called a sporadic task
- Aperiodic task: When the recurrence is random but there is no minimum time difference between two jobs of the same task, means they may even appear at the same time, the task is called an aperiodic task

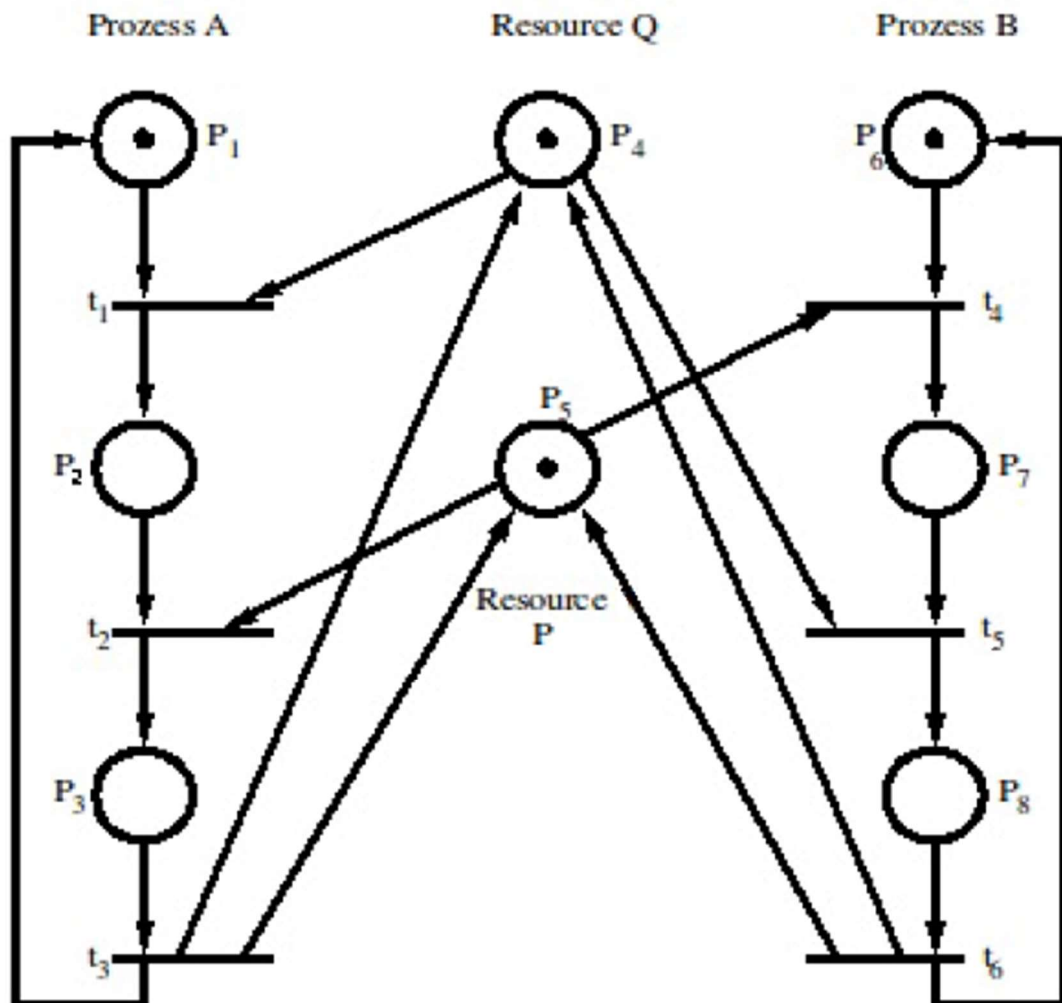
Task 3: EDF Task to Complete

T1: (1,7), T2: (2,10), T3: (3,15)

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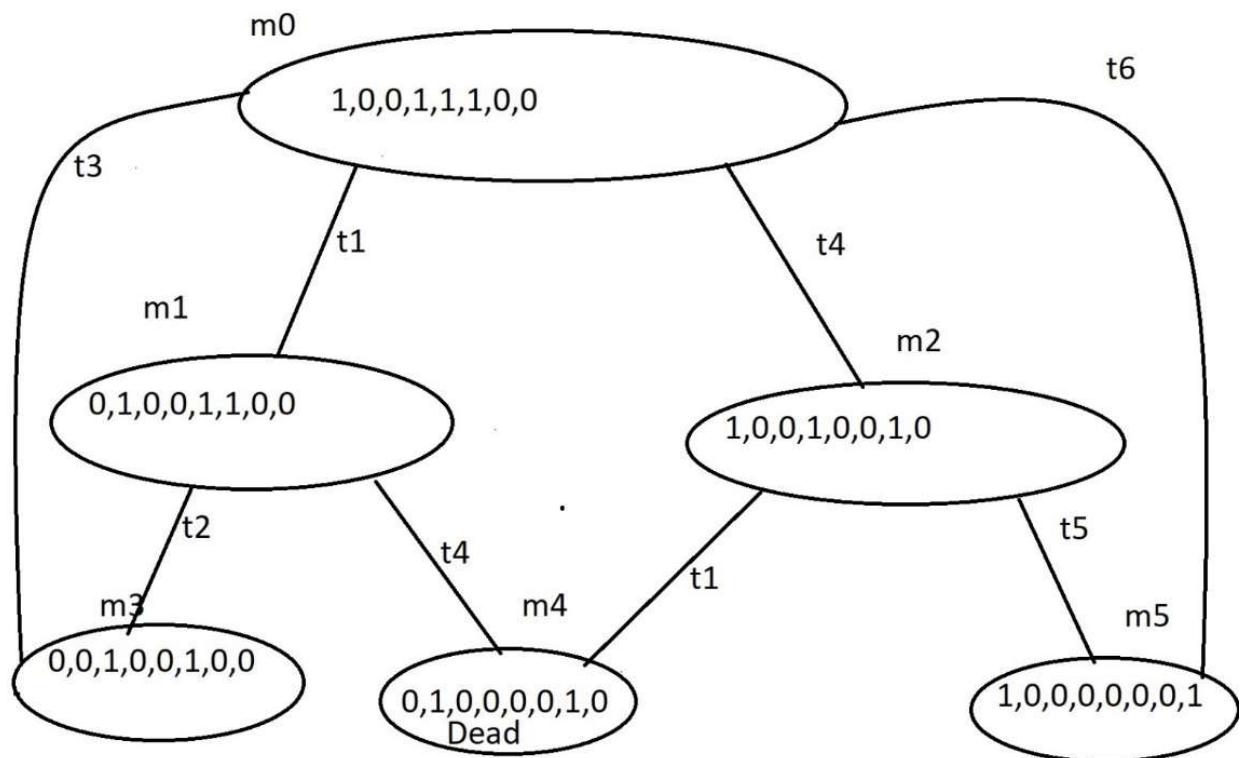


Task 4: Petri Nets: Make a reachability Graph out of given Petri Nets



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Solution:



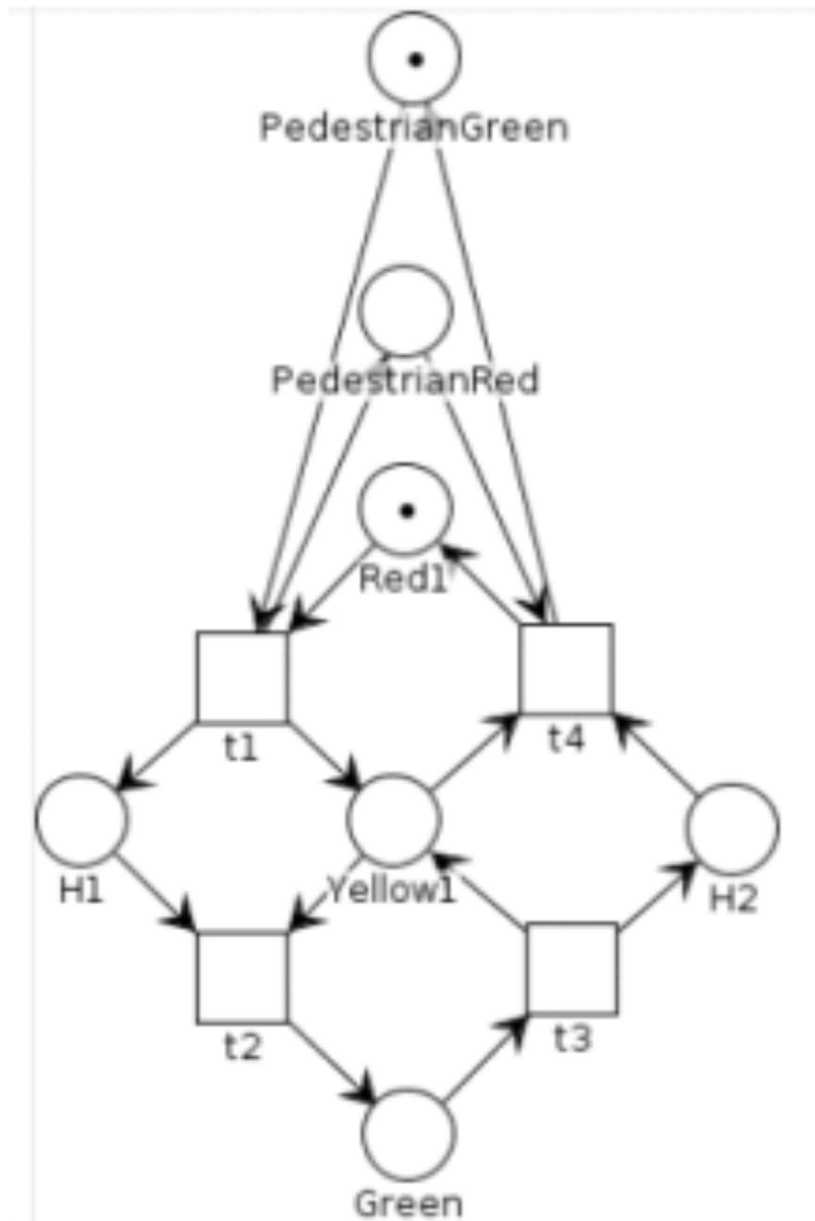
Task 5a: Please model a Petri-Net that represents the different statuses of traffic lights for pedestrians. It has two states: <red> and <green>.

Task 5b: b) Please model a Petri-Net that represents the different statuses of traffic lights for automotivetraffic. It has three states: <red>, <yellow> and <green>. The switching sequence is <red> to <yellow> to <green> to <yellow> to <red>. There is one color only at a time.

Task 5c: Combine the two Petri-Nets of a) and b) in that way that pedestrians have <green> only when the automative traffic has <red>. (Make them mutual exclusive).

- Combined Solution!!

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Task 6: RMS Scheduling

a)

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Aufgabe 6a (in Zusammenhang mit Nr. 5, aber auch ohne Nr. 5 lösbar):

Wenn Sie Einheiten von 1/24 Sekunden verwenden, ergeben sich folgende Parameter:

$p_{\text{foot pedal}}$	= 12
p_{wheel}	= 3
$p_{\text{heartbeat}}$	= 8

Die Rechenzeiten (execution times) für die Tasks sind gegeben als:

$t_{\text{exec:foot pedal}}$	= 2
$t_{\text{exec:wheel}}$	= 1
$t_{\text{exec:heartbeat}}$	= 3

Damit ergibt sich: $T_i = (p_i, t_{i,\text{exec}})$

$T_{\text{foot pedal}} = (12, 2); T_{\text{wheel}} = (3, 1); T_{\text{heartbeat}} = (8, 3)$

Bitte berechnen Sie die CPU-Auslastung für alle Tasks und in Summe (geben Sie den Rechenweg und nicht nur das Ergebnis an).

Was bedeutet dies für die Schedulability (Ja/Nein/Vielleicht) und warum)?

n ist the Anzahl der periodic tasks (texe) (in der Aufgabe ist n=3)

Formel: $U = \text{texc}/P$ und am ende alle addieren

Test durchführen

$$U \leq n * (2^{1/n} - 1)$$

Wenn:

Load $U = 1$ means processor never idles

Load $U > 1$ there is no feasible schedule

Load $U < 1$ means this test does not exclude that feasible schedule may exist^

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b)

RMS

Load of a task: $U_i = \frac{e_i}{D_i}$ (e_i : execution time / D_i : until expiration)

System load: $U = \sum_{i=1}^m U_i = \sum_{i=1}^m \frac{e_i}{D_i}$

Task1: $\frac{7}{24} + \frac{1}{8} + \frac{4}{14} + \frac{3}{15} = 0,90238$ this test does not exclude that a feasible schedule exists

Task2: $\frac{6}{20} + \frac{3}{12} + \frac{4}{10} = 0,95$ this test does not exclude that a feasible schedule exists

Task3: $\frac{7}{20} + \frac{3}{12} + \frac{4}{10} = 1$ processor never idles

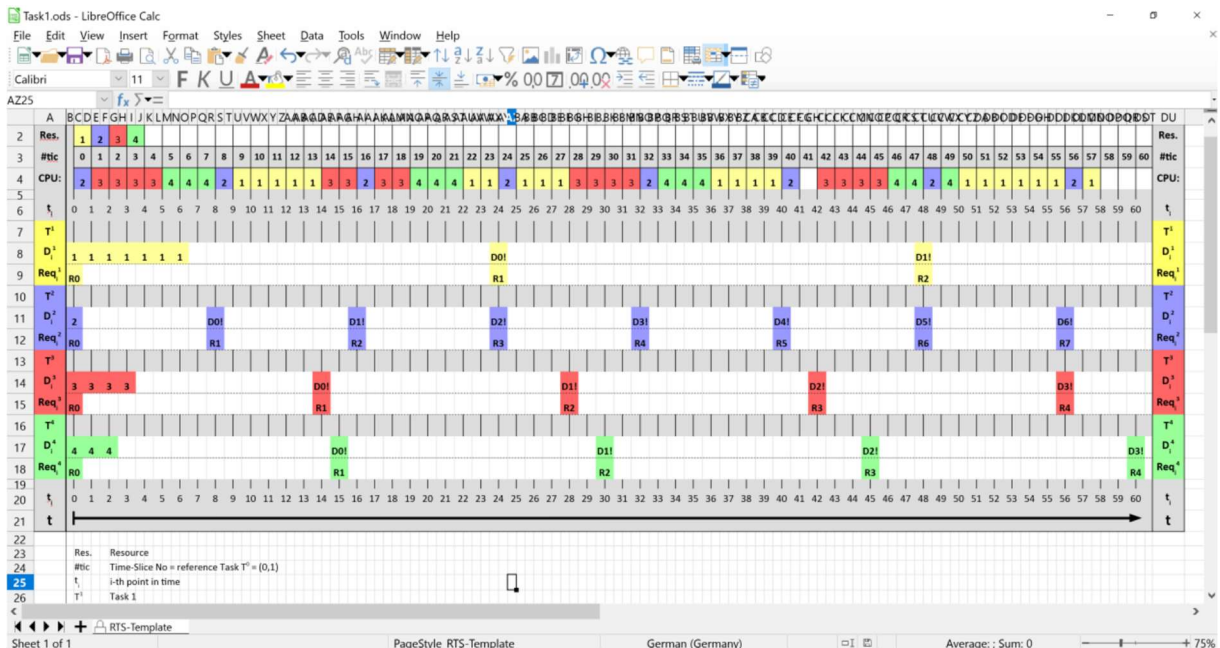
Schedulability test: $U \leq m * (2^{\frac{1}{m}} - 1)$ (m is the number of periodic tasks)

Task1: $U \nless 4 * (2^{\frac{1}{4}} - 1) = 0,7568$ ($U = 0,90238$)

Task2: $U \nless 3 * (2^{\frac{1}{3}} - 1) = 0,78$ ($U = 0,95$)

Task3: $U \nless 3 * (2^{\frac{1}{3}} - 1) = 0,78$ ($U = 1$)

c) RMS Calculation



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Task 7: Microcontroller

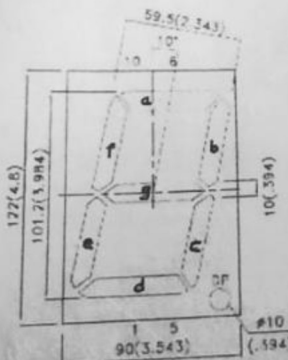
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Exercise 7: Microcontroller (8 Punkte)
Sie sollen über einen Microcontroller (z.B. STR9) eine 7-Segment-Anzeige ansteuern.
Die Spezifikation der 7-Segment-Anzeige, die mit den Microcontroller-Ports verbunden ist, ist folgende:

Die 7-Segment-Anzeige ist mit den folgenden Ports verbunden:

- Segment g an GPIO 4.0,
- Segment f an GPIO 4.1,
- Segment e an GPIO 4.2,
- Segment d an GPIO 4.3,
- Segment c an GPIO 4.4,
- Segment b an GPIO 4.5,
- Segment a an GPIO 4.6.

- Die Segmente sind "active low" anzusteuern.
- Sie können alle Bits für GPIO4 folgendermaßen setzen: $\text{GPIO4} \rightarrow \text{DATA}[255:2] = \text{BITARRAY}$;
z.B. um eine "1" anzuzeigen: $1100\ 1111 = 0x\text{CF} \rightarrow \text{GPIO4} \rightarrow \text{DATA}[255:2] = 0x\text{CF}$;



Welche hexadezimal Zahl müssen Sie speichern um eine "4" anzuzeigen?
 $\text{GPIO4} \rightarrow \text{DATA}[255:2] = 0x\text{_____}$;

Welche hexadezimal Zahl müssen Sie speichern um eine "2" anzuzeigen?
 $\text{GPIO4} \rightarrow \text{DATA}[255:2] = 0x\text{_____}$;

1 = off 0 = on. Lösung durch Formel $1(-) \times (a) \times (b) \times (c) \times (d) \times (e) \times (f) \times (g)$

4 = 1100 1100

2 = 1001 0010

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Task 8: Fahrrad Quiz:

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Aufgabe 5: Fahrrad (3 Punkte)



Die Rate des Tretlagers, die der Fahrer an den Pedalen treibt, beträgt zwischen 0 und ^{max}150 Umdrehungen pro Minute. ^{60 sec}

Die Übersetzung der Gangschaltung ist 52 Zähne (vorne) zu 13 Zähnen (hinten) = 4:1;

Das bedeutet, dass sich das Hinterrad viermal so schnell dreht wie das Tretlager(Pedale).

Die Rate des Herzschlages des Fahrers liegt zwischen 60 und ^{max}180 Schlägen pro Minute. ^{60 sec}

Ihre Aufgabe ist die Entwicklung eines Fahrradcomputers, welcher folgende Raten ermittelt und anzeigt:

1. Tretlager/Pedale (foot pedal), 2. Hinterrad (wheel), 3. Herzschlag des Fahrers (heartbeat).

Bitte berechnen Sie die minimale Periode [in Sekunden] für die drei Parameter:

$$\begin{aligned} \text{min: } p_{\text{foot pedal}} &= \frac{1}{\text{Rate}} = \frac{1}{150 \cdot 60} = 0.4 \\ \text{min: } p_{\text{wheel}} &= 0.4 \times 4 = 1.6 \\ \text{min: } p_{\text{heartbeat}} &= \frac{1}{180 \cdot 60} = 0.33 \end{aligned}$$

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Task 9a: Classification of Real-Time Systems:

1. Consequence of missing deadline
2. Reliability and fault tolerance
3. Distribution: centralized or distributed RTS
4. Interactive or autonomous system
5. Hierarchical or independent systems
6. Time-driven or event-driven RTS
7. Periodic, aperiodic, sporadic RTS
8. Cyclic or asynchronous scheduling

Aufgabe 9b: Pagefault (Minor Pagefault):

A page fault is a type of exception raised by computer hardware when a running program accesses a memory page that is not currently mapped by the memory management unit (MMU) into the virtual address space of a process.

If the page is loaded in memory at the time the fault is generated, but is not marked in the memory management unit as being loaded in memory, then it is called a minor or soft page fault. No I/O operations are necessary.